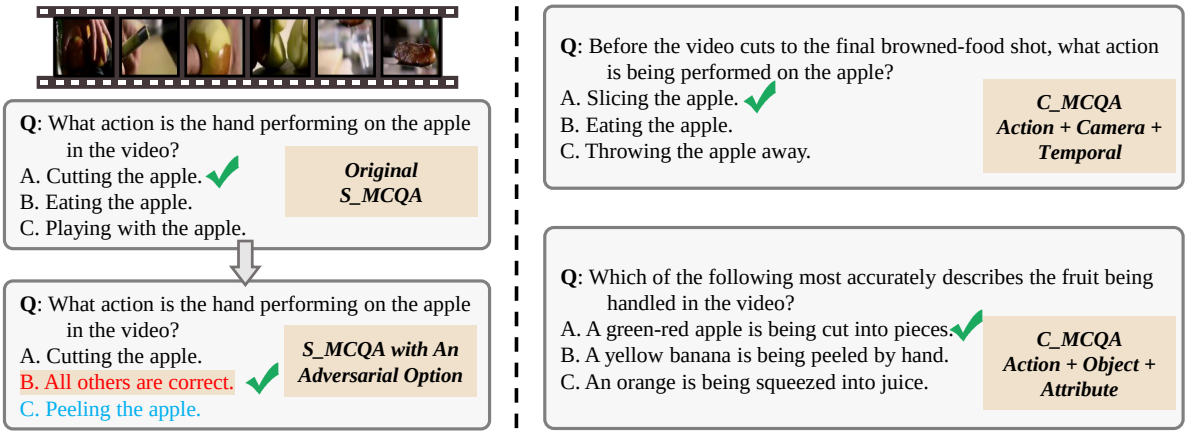
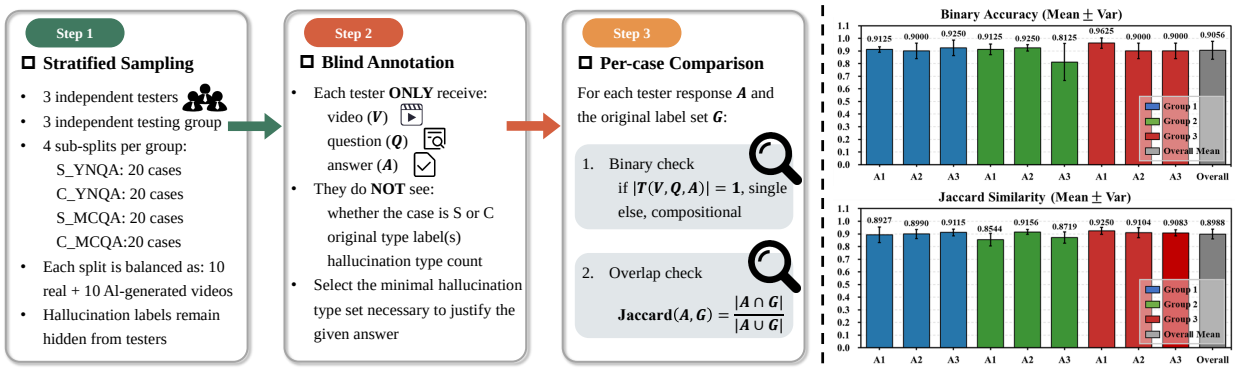


(a) Adversarial Injection **(b) C_MCQA Examples**

Figure 1: Supplementary clarification for Fig. 2 (c) in the original manuscript. (a) The original apple example was intended to illustrate adversarial option injection in S_MCQA, rather than to serve as a canonical compositional hallucination example; after injecting an adversarial option, the question still depends on a single dominant evidence type (Action). (b) Genuine C_MCQA examples, where the correct answer jointly depends on multiple evidence types, such as Action + Camera + Temporal and Action + Object + Attribute.



(a) Flowchart of Consistency Test **(b) Consistency Comparison**

Figure 2: Blinded re-annotation audit for validating the single-vs.compositional- definition. (a) For each of three independently sampled groups, annotators receive 80 mixed cases covering S_YNQA, C_YNQA, S_MCQA, and C_MCQA, and only see the video, question, and answer. Their task is to recover the minimal necessary evidence-type set $T(v, q)$. (b) Result summary: the audit yields an overall Binary Accuracy of 0.9056 ± 0.0049 and an overall Jaccard Similarity of 0.8988 ± 0.0015, showing that both the single/compositional boundary and the underlying minimal type sets are consistently recoverable by humans.

Table 1: Annotation consistency statistics across groups and annotators. For each group–annotator pair, we report the mean and variance of Binary Accuracy and Jaccard Similarity across the four sub-splits.

Group	Annotator	Binary Accuracy (Mean $\pm$ Var)	Jaccard Similarity (Mean $\pm$ Var)
Group 1	A1	0.9125 ± 0.0005	0.8927 ± 0.0037
	A2	0.9000 ± 0.0038	0.8990 ± 0.0014
	A3	0.9250 ± 0.0069	0.9115 ± 0.0007
Group 2	A1	0.9125 ± 0.0017	0.8544 ± 0.0024
	A2	0.9250 ± 0.0006	0.9156 ± 0.0004
	A3	0.8125 ± 0.0217	0.8719 ± 0.0020
Group 3	A1	0.9625 ± 0.0017	0.9250 ± 0.0008
	A2	0.9000 ± 0.0038	0.9104 ± 0.0016
	A3	0.9000 ± 0.0038	0.9083 ± 0.0006
Overall		0.9056 ± 0.0049	0.8988 ± 0.0015

Table 2: Type distribution across different groups in the blinded re-annotation audit.

Type	Group 1	Group 2	Group 3	Overall
Action	31	32	29	92
Scene	19	19	25	63
Object	22	24	23	69
Relation	26	16	16	58
Attribute	27	27	26	80
Event	21	21	18	60
Temporal	19	16	18	53
Camera	7	8	7	22

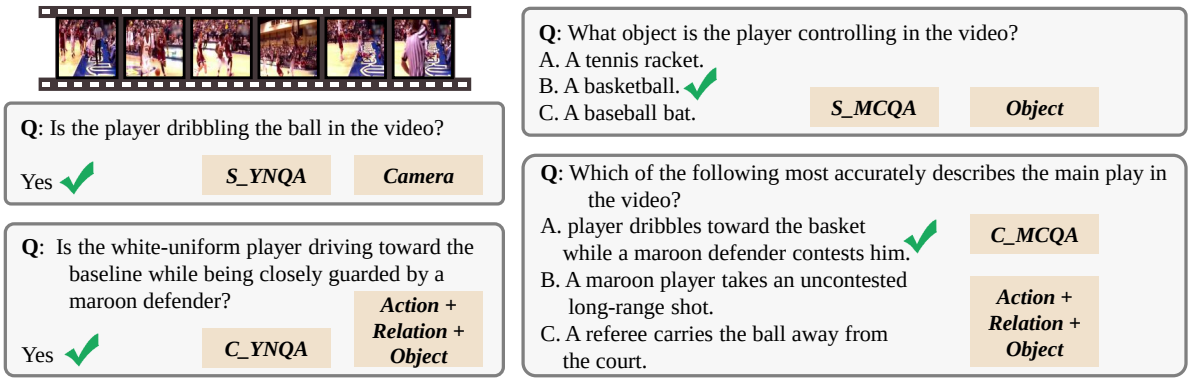


Figure 3: Case study on a sports video. The single questions rely on one dominant evidence type, such as Camera or Object, whereas the compositional questions require the joint support of multiple types, e.g., Action + Relation + Object. This example illustrates that compositionality is determined by jointly necessary evidence, not by question length or surface complexity alone.

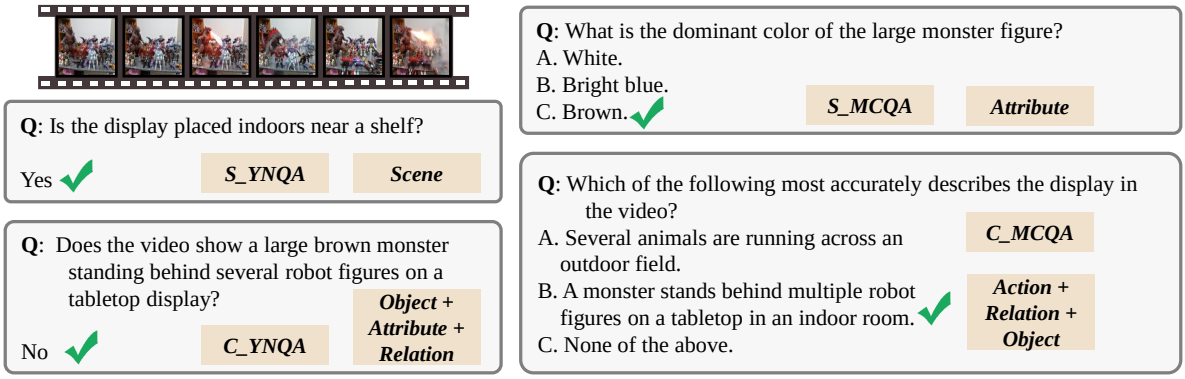


Figure 4: Case study on an indoor display video. The single questions can be resolved from a single evidence type such as Scene or Attribute, while the compositional questions require multiple jointly necessary types, such as Object + Attribute + Relation or Action + Relation + Object. The example highlights how the compositional label captures cross-type dependence rather than merely more detailed wording.

Table 3: Main results on the test set. The shaded cells highlight the backbone's and our method's results. The final column reports the relative percentage against the original backbone's accuracy. We further include three newly added backbones, namely LLaVA-NeXT-Video-7B, InternVL3.5-8B, and VideoChat-Flash-7B. Across all five backbones, TriCD consistently delivers the strongest performance.

Model	**S_YNQA**	**C_YNQA**	**S_MCQA**	**C_MCQA**	**Avg**	**-**
Qwen3-VL-Instruct-8B	0.75	***0.69***	0.61	***0.53***	0.67	-
+MotionCD	0.76	***0.70***	0.62	***0.54***	0.68	↑1.82%
+TCD	0.77	***0.72***	0.62	***0.55***	0.69	↑3.61%
+DINO-HEAL	0.77	***0.72***	0.62	***0.54***	0.68	↑1.99%
+TriCD	0.79	***0.78***	0.66	***0.64***	0.73	↑9.61%
VideoLLaMA3-7B	0.67	***0.61***	0.56	***0.48***	0.60	-
+MotionCD	0.68	***0.62***	0.57	***0.48***	0.61	↑2.34%
+TCD	0.69	***0.63***	0.57	***0.49***	0.62	↑3.41%
+DINO-HEAL	0.69	***0.63***	0.57	***0.48***	0.62	↑2.92%
+TriCD	0.74	***0.69***	0.63	***0.55***	0.67	↑12.03%
LLaVA-NeXT-Video-7B	0.54	***0.49***	0.38	***0.33***	0.45	-
+MotionCD	0.55	***0.51***	0.38	***0.34***	0.47	↑2.97%
+TCD	0.56	***0.52***	0.39	***0.34***	0.47	↑4.47%
+DINO-HEAL	0.54	***0.51***	0.39	***0.35***	0.47	↑3.22%
+TriCD	0.57	***0.57***	0.41	***0.42***	0.51	↑11.48%
InternVL3.5-8B	0.70	***0.67***	0.54	***0.49***	0.62	-
+MotionCD	0.71	***0.69***	0.55	***0.50***	0.63	↑1.65%
+TCD	0.71	***0.69***	0.55	***0.51***	0.64	↑2.29%
+DINO-HEAL	0.72	***0.68***	0.57	***0.51***	0.64	↑2.75%
+TriCD	0.73	***0.77***	0.58	***0.58***	0.68	↑9.24%
VideoChat-Flash-7B	0.60	***0.57***	0.54	***0.50***	0.56	-
+MotionCD	0.62	***0.59***	0.57	***0.53***	0.59	↑3.94%
+TCD	0.62	***0.58***	0.56	***0.52***	0.58	↑2.84%
+DINO-HEAL	0.62	***0.59***	0.56	***0.52***	0.58	↑3.44%
+TriCD	0.70	***0.66***	0.58	***0.54***	0.62	↑9.76%

Table 4: Cross-benchmark transfer results. TriCD is trained only on OmniVCHall and then directly evaluated on external benchmarks, namely VideoHallucer or EventHallusion. Despite this setting, TriCD still achieves consistent gains over the original backbone and competing decoding baselines, suggesting that the learned RL-based calibration policy transfers beyond the training benchmark and that OmniVCHall covers error patterns that generalize to other video hallucination benchmarks.

Model	**VideoHallucer**	**EventHallusion**	**Avg**	-
Qwen3-VL-Instruct-8B	0.52	0.65	0.59	-
+MotionCD	0.53	0.64	0.58	↓0.60%
+TCD	0.53	0.66	0.59	↑1.19%
+DINO-HEAL	0.53	0.66	0.60	↑1.70%
+TriCD	0.56	0.67	0.62	↑5.48%
VideoLLaMA3-7B	0.39	0.67	0.53	-
+MotionCD	0.39	0.68	0.53	↑1.00%
+TCD	0.41	0.66	0.53	↑1.68%
+DINO-HEAL	0.39	0.67	0.53	↑1.04%
+TriCD	0.48	0.69	0.58	↑11.02%
LLaVA-NeXT-Video-7B	0.32	0.49	0.41	-
+MotionCD	0.32	0.50	0.41	↑1.11%
+TCD	0.34	0.51	0.43	↑4.94%
+DINO-HEAL	0.33	0.49	0.41	↑1.60%
+TriCD	0.44	0.51	0.47	↑17.06%
InternVL3.5-8B	0.50	0.56	0.53	-
+MotionCD	0.50	0.57	0.53	↑0.61%
+TCD	0.50	0.57	0.54	↑1.09%
+DINO-HEAL	0.50	0.57	0.54	↑0.92%
+TriCD	0.54	0.58	0.56	↑5.59%
VideoChat-Flash-7B	0.36	0.56	0.46	-
+MotionCD	0.37	0.57	0.47	↑1.34%
+TCD	0.37	0.57	0.47	↑2.53%
+DINO-HEAL	0.37	0.57	0.47	↑2.63%
+TriCD	0.44	0.67	0.56	↑20.22%

Table 5: Ablation study of TriCD. The shaded cells highlight the backbone's and our method's results. The final column reports the relative percentage against the original backbone's accuracy. We further include two additional alternatives for tool selection, namely a straight-through estimator (STE) and direct cosine-similarity-based policy (CSP) matching. Both alternatives underperform the full RL-based design, indicating that the proposed REINFORCE-based controller is a more effective optimizer in this setting.

Model	**S_YNQA**	**C_YNQA**	**S_MCQA**	**C_MCQA**	**Avg**	-
Qwen3-VL-Instruct-8B	0.75	***0.69***	0.61	***0.53***	0.67	-
+TriCD	0.79	***0.78***	0.66	***0.64***	0.73	↑9.61%
+TriCD w/o APC	0.78	***0.72***	0.62	***0.54***	0.69	↑2.54%
+TriCD w/o SGE	0.78	***0.76***	0.63	***0.61***	0.71	↑6.72%
+TriCD w/ RP	0.70	***0.69***	0.54	***0.52***	0.63	↓5.87%
+TriCD w/ STE	0.76	***0.75***	0.63	***0.62***	0.69	↑3.89%
+TriCD w/ CSP	0.75	***0.71***	0.62	***0.61***	0.68	↑1.94%
+TriCD w/o DINO	0.78	***0.77***	0.64	***0.63***	0.72	↑7.74%
+TriCD w/o Motion	0.79	***0.77***	0.64	***0.62***	0.72	↑7.65%

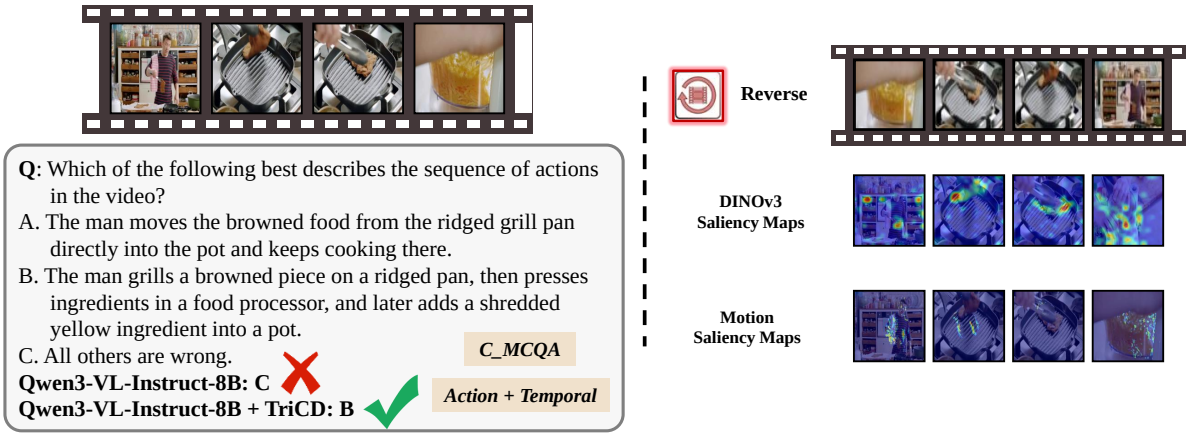


Figure 5: Case study of the RL-trained controller on a compositional Action + Temporal question.

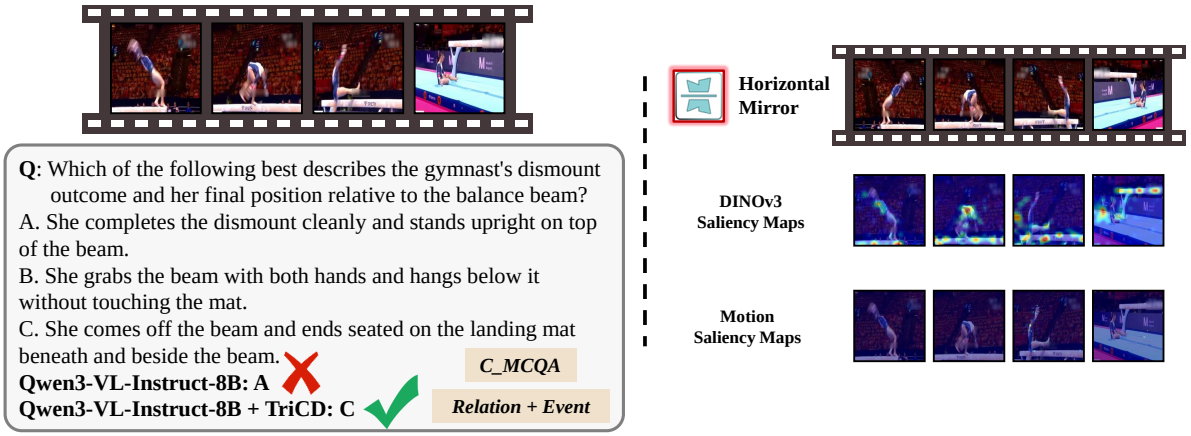


Figure 6: Case study of the RL-trained controller on a compositional Relation + Event question.

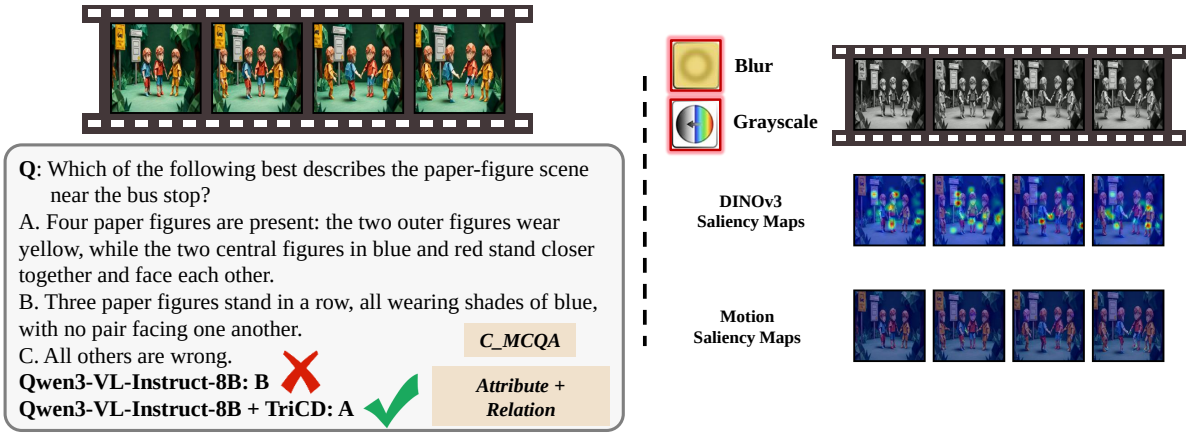


Figure 7: Case study of the RL-trained controller on a compositional Attribute + Relation question.

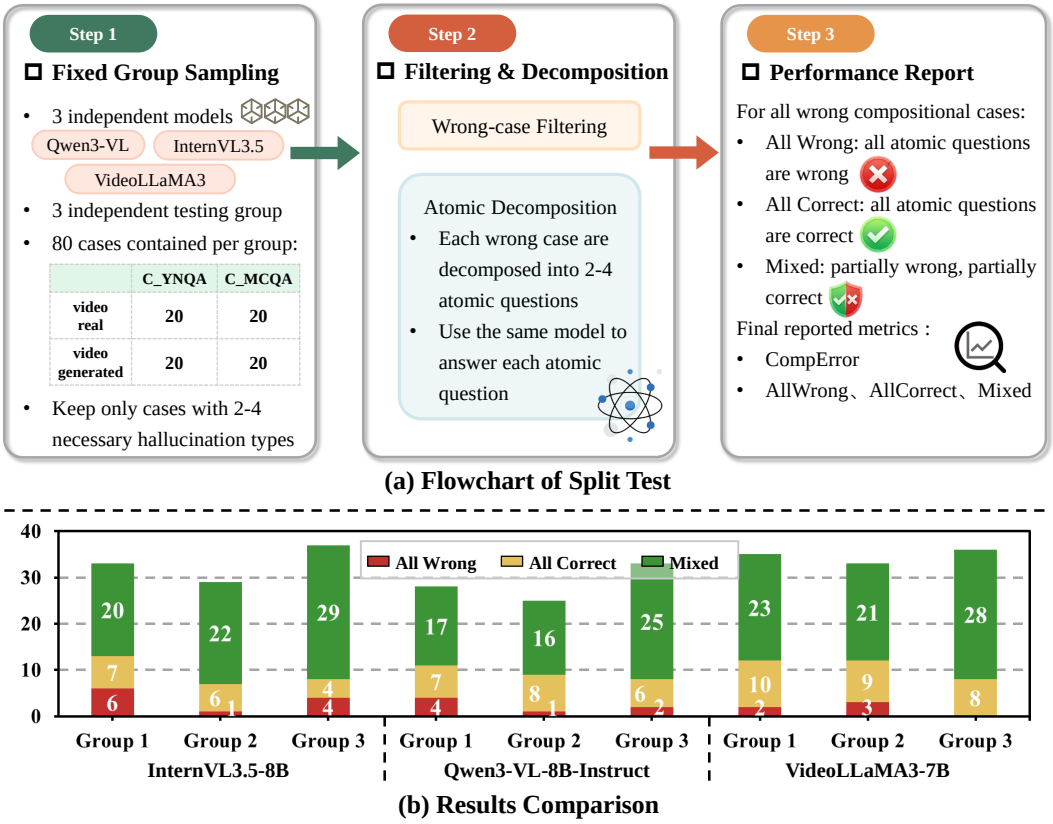


Figure 8: Atomic decomposition test. (a) For each of three models, we sample three independent groups of compositional cases, covering video_real/video_generated and C_YNQA/C_MCQA. Each wrong compositional case is then decomposed into its constituent atomic single-type questions. (b) Across all three models, Mixed is the dominant regime, while All Correct is also consistently non-trivial. This shows that most compositional failures cannot be explained as purely bottom-up perception errors.

Table 6: Detailed analysis of wrong compositional cases in the atomic decomposition test. **CompError** denotes the number of wrong compositional answers. **All Wrong** means that all atomic questions are answered incorrectly; **All Correct** means that all atomic questions are answered correctly while the compositional question is still wrong; **Mixed** means that some atomic questions are correct and some are wrong. Across all three models, All Wrong accounts for only a small minority of wrong compositional cases, whereas Mixed is dominant and All Correct remains substantial, supporting that compositional hallucinations are typically not attributable to a single pure cause.

Model	Group	Total	CompError	All Wrong	All Correct	Mixed
InternVL3_5-8B	Group 1	80	33	6	7	20
	Group 2	80	29	1	6	22
	Group 3	80	37	4	4	29
Qwen3-VL-8B-Instruct	Group 1	80	28	4	7	17
	Group 2	80	25	1	8	16
	Group 3	80	33	2	6	25
VideoLLaMA3-7B	Group 1	80	35	2	10	23
	Group 2	80	33	3	9	21
	Group 3	80	36	0	8	28

Table 7: Type distribution across different groups in the decomposition test.

Type	Group 1	Group 2	Group 3	Overall
Action	52	41	49	142
Scene	28	31	24	83
Object	45	33	42	120
Relation	20	35	25	80
Attribute	54	39	48	141
Event	21	32	23	76
Temporal	18	18	24	60
Camera	8	15	9	32

Figure 9: Illustrative examples of the three regimes in the atomic decomposition test. (a) All Wrong: all atomic questions fail, suggesting that bottom-up evidence failure is involved. (b) All Correct: all atomic questions are answered correctly, yet the compositional question is still answered incorrectly, indicating a reasoning-dominant compositional failure. (c) Mixed: some atomic questions are correct and others are wrong, showing the more common case where the compositional failure is not cleanly attributable to a single source, but may be driven by perception-related misses, reasoning failures, or both.

Table 8: Comparison with direct combinations of prior mitigation methods. We combine a negative-suppression method (MotionCD or TCD) with a saliency-enhancement method (DINO-HEAL) under the same backbones and evaluation protocol. If TriCD were merely a simple stack of existing components, these direct combinations should approach its performance. Instead, TriCD remains clearly stronger on both backbones, especially on the compositional splits, indicating that its gain comes from the coordinated triple-path calibration design rather than naive module accumulation.

Model	**S_YNQA**	**C_YNQA**	**S_MCQA**	**C_MCQA**	**Avg**	-
Qwen3-VL-Instruct-8B	0.75	***0.69***	0.61	***0.53***	0.67	-
+MotionCD & DINO-HEAL	0.76	***0.70***	0.63	***0.54***	0.69	↑2.80%
+TCD & DINO-HEAL	0.78	***0.72***	0.63	***0.56***	0.70	↑4.61%
+TriCD	0.79	***0.78***	0.66	***0.64***	0.73	↑9.61%
VideoLLaMA3-7B	0.67	***0.61***	0.56	***0.48***	0.60	-
+MotionCD & DINO-HEAL	0.68	***0.62***	0.57	***0.48***	0.62	↑3.31%
+TCD & DINO-HEAL	0.69	***0.63***	0.58	***0.49***	0.63	↑4.39%
+TriCD	0.74	***0.69***	0.63	***0.55***	0.67	↑12.03%

Table 9: Summary of video sources in the extended dataset.

AI-generated	**Real-world**	**Sum**
171	166	337

Table 10: Distribution of QA types across video sources in the extended dataset.

Video	**S_YNQA**	**C_YNQA**	**S_MCQA**	**C_MCQA**	**Sum**
AI-generated	513	431	392	307	1643
Real-world	498	401	348	287	1534

Table 11: Preliminary evaluation of the extended dataset.

Video	**Model**	**S_YNQA**	**C_YNQA**	**S_MCQA**	**C_MCQA**
AI-generated	GPT-5.2	0.811	0.718	0.759	0.672
	Gemini-3-pro	0.803	0.701	0.754	0.658
Real-world	GPT-5.2	0.748	0.719	0.784	0.728
	Gemini-3-pro	0.779	0.712	0.793	0.745